



NEO-THERMO: A LOW-COST ADD-ON FOR WARMING NEONATAL RESPIRATORY GAS

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Need Statement

NICU clinicians in resource-limited hospitals need a low-cost, portable, and locally maintainable accessory that monitors and warms respiratory gas delivered to newborns while working with existing neonatal breathing equipment.

Background and Motivation

- Uganda's neonatal mortality rate remained approximately 27 deaths per 1,000 live births between (Kananura et al. 2016)
- Cold, dry respiratory gas can increase heat and moisture loss from the airways of premature infants during respiratory support (PIPER Neonatal Education, 2013).
- Inadequate humidification can thicken airway secretions and increase airway resistance and work of breathing (PIPER Neonatal Education, 2013).
- Neonatal guidance recommends at least 34°C with appropriate humidification during CPAP and approximately 37°C for intubated infants (Kadam, n.d.).
- At the partner hospital, the humidifier's heating component was reported to be broken and unused, leaving respiratory gas without active warming (BioSpark Stakeholder Interviews, 2026).

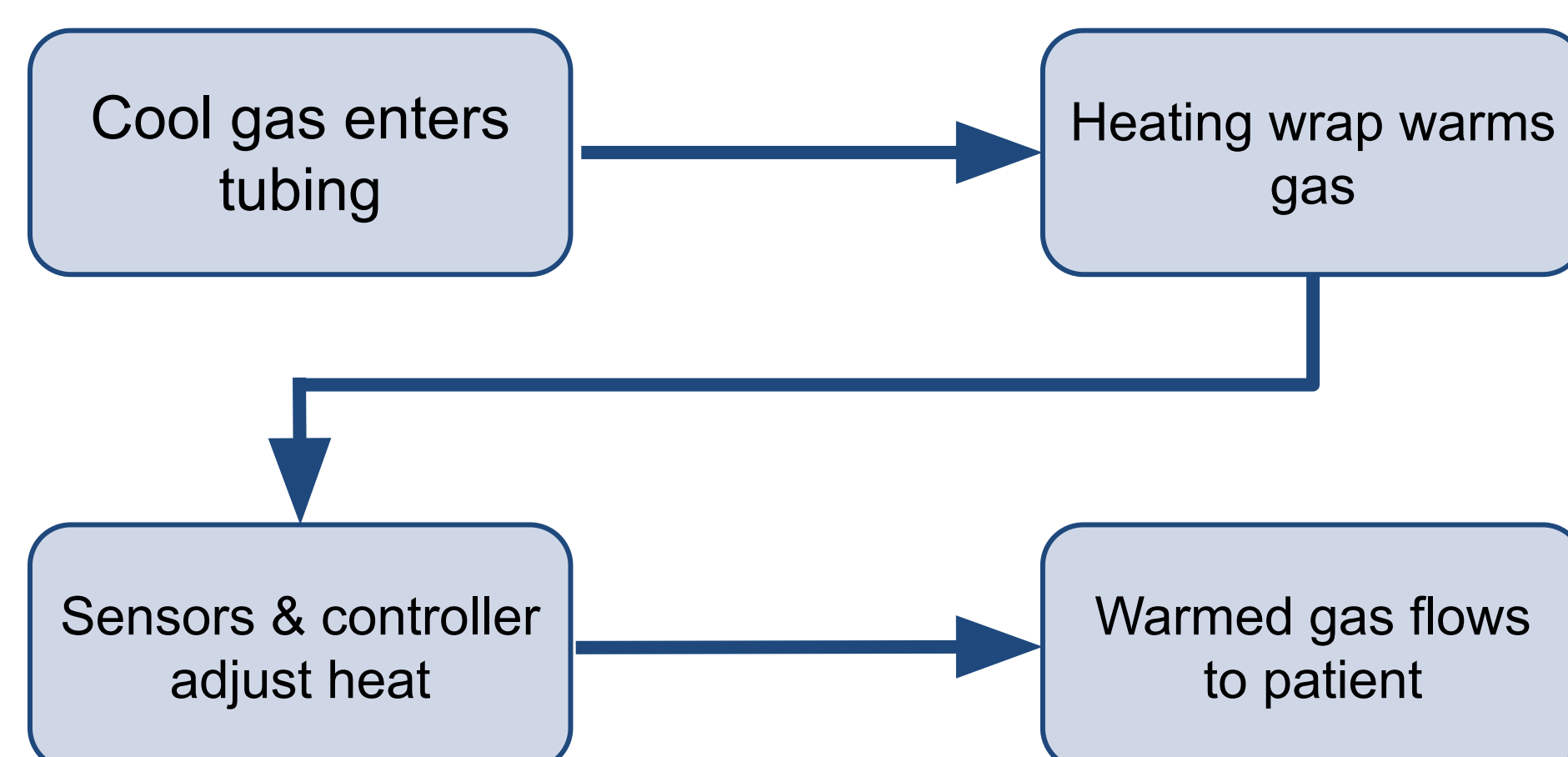
Our Solution

Neo-Thermo addresses the lack of active respiratory-gas warming by adding a warming stage near the patient. Warming the gas shortly before delivery limits the distance over which it can cool before reaching the infant.

Rather than replacing the hospital's oxygen-delivery equipment, Neo-Thermo is designed as an add-on to existing respiratory tubing. This retrofit approach makes gas warming more practical for resource-limited facilities where replacing an entire respiratory system may be difficult.

How It Works

Neo-Thermo wraps around breathing tubing near the patient and warms the gas through the tubing wall without contacting the airflow. Two external temperature sensors help estimate the gas temperature inside the tube. The system automatically controls the heater to maintain the target range, while a display, colored lights, and an alarm communicate temperature and potential faults.



Prototype Design

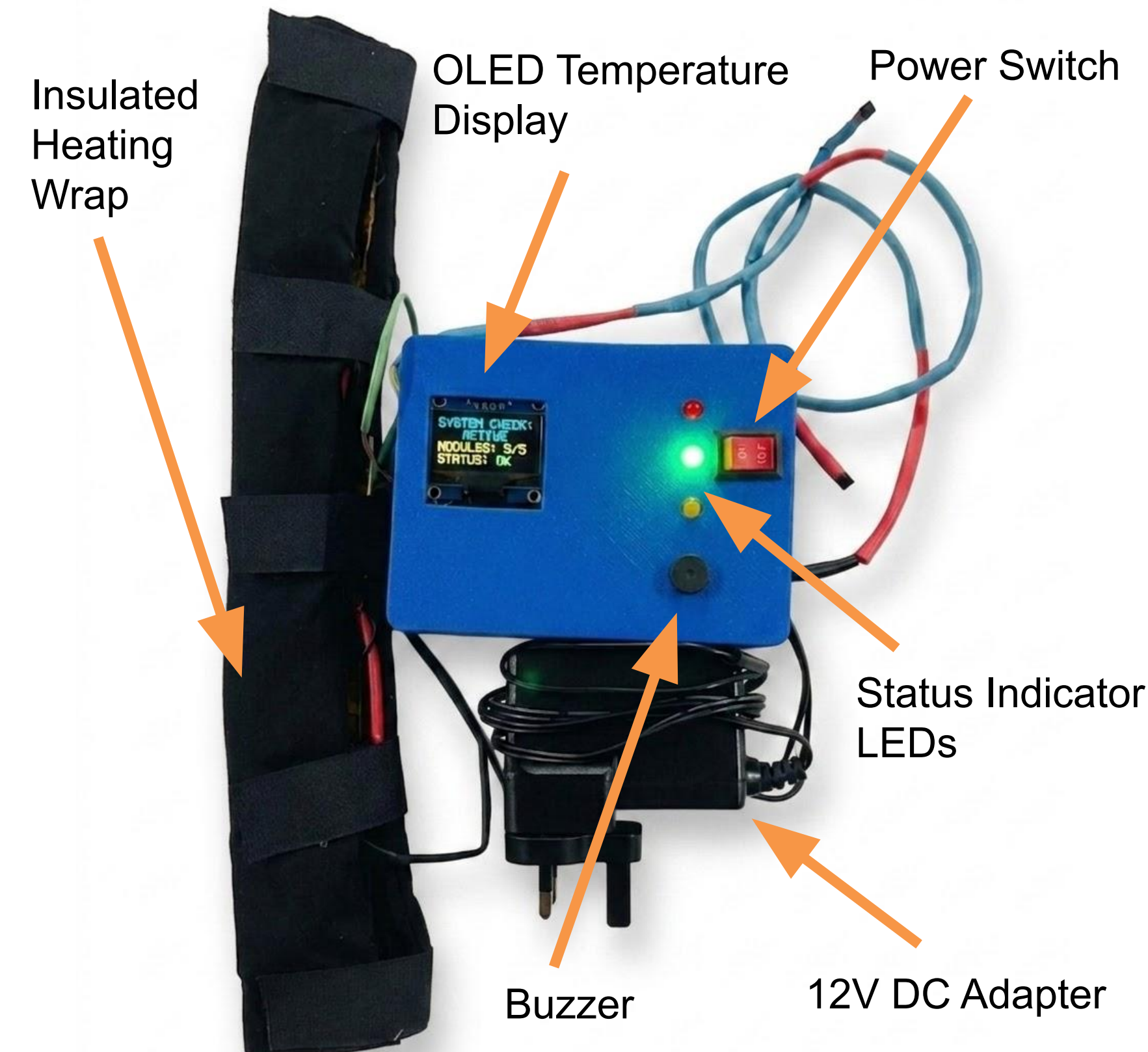


Figure 1. Neo-Thermo prototype and components.

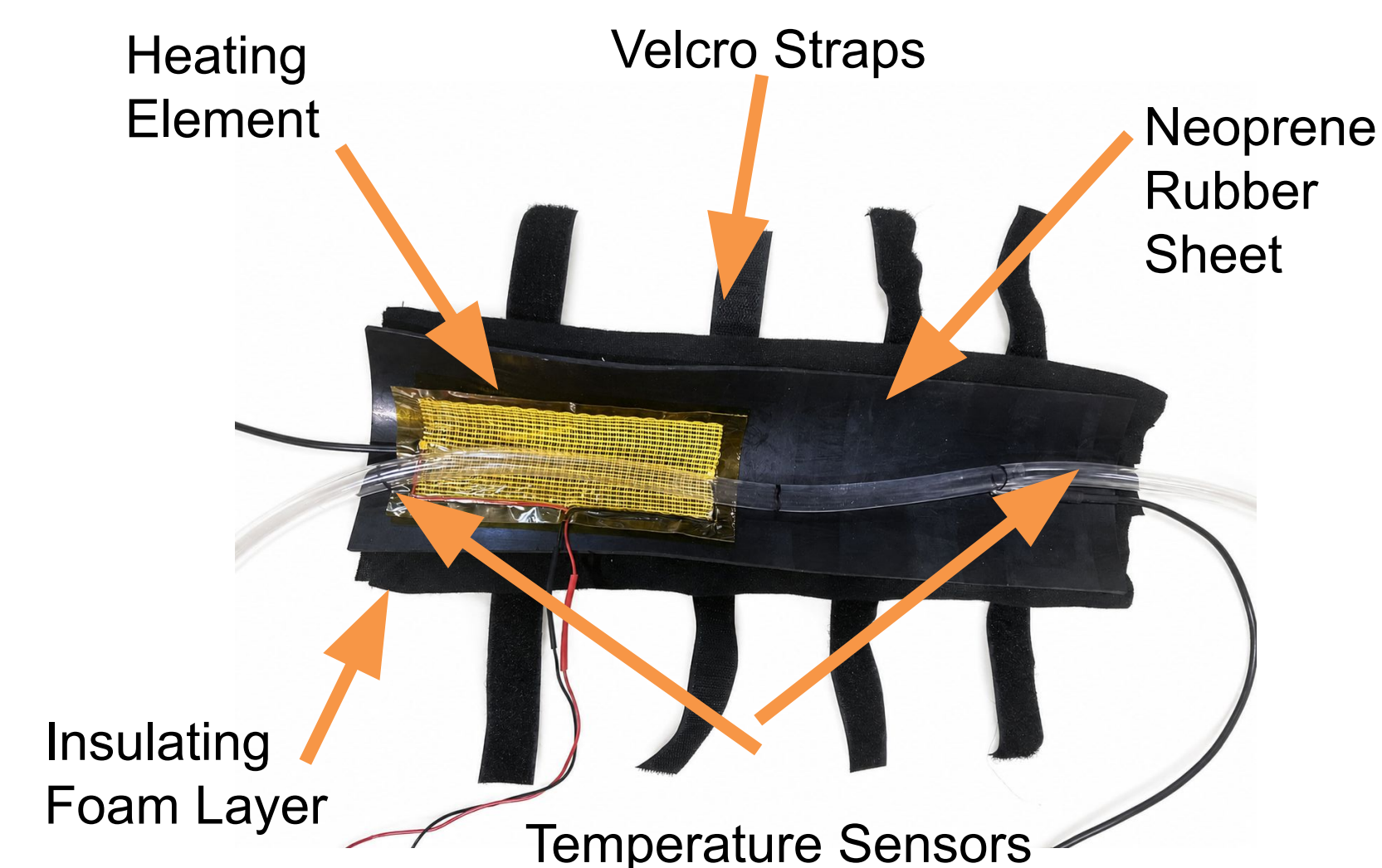


Figure 2. Heating-wrap construction.

Internal Temperature Estimation

To keep sensing components outside the airflow, Neo-Thermo estimates internal gas temperature from heater and outer-tubing measurements. In a separate model-evaluation run, estimates were compared with direct internal measurements using a $\pm 1^\circ\text{C}$ criterion.

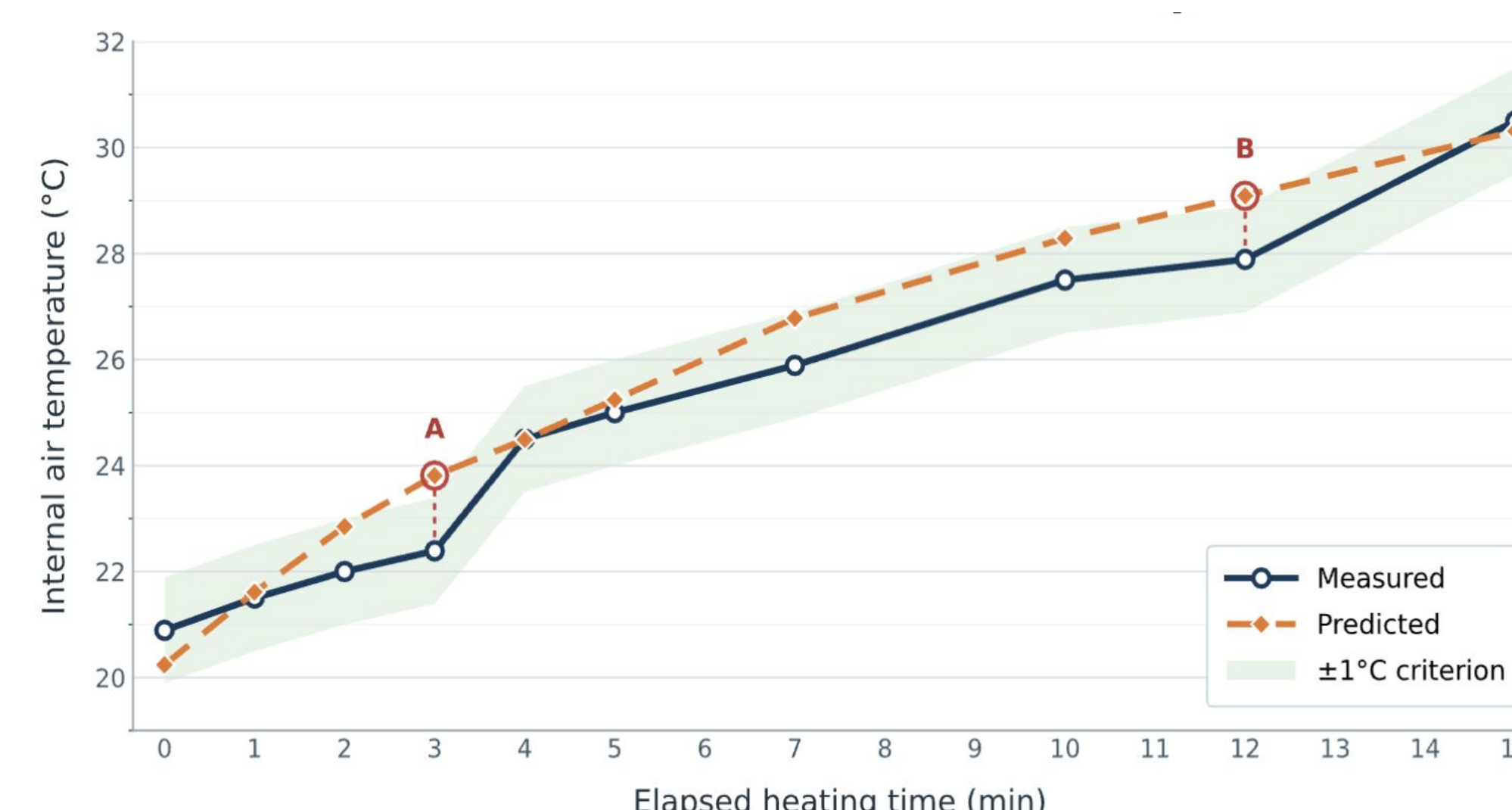


Figure 3. Predicted and measured internal-gas temperature. Eight of 10 estimates met the $\pm 1^\circ\text{C}$ criterion (mean absolute error = 0.64°C); points A and B exceeded it.

Design Objectives and Testing

Category	Criteria	Test Description	Results
Warming Performance	Outlet gas reaches and remains between 34–37°C	Recorded outlet-gas temperature every 30 seconds during dry-air testing.	PASS
Internal Temperature Estimation Accuracy	Estimated temperatures remain within $\pm 1^\circ\text{C}$ of direct measurements	Compared estimated gas temperature with measurements taken inside the tubing.	PARTIAL: MAE 0.64°C
Physical Compatibility	Fits tested neonatal breathing equipment without permanent modification	Device was connected ventilator and Leve CPAP breathing circuits to confirm fit and air flow without modification	PASS
Affordability	Prototype component cost does not exceed \$70	Added the documented cost of all prototype components.	PASS - \$65
Ease of Use	An unfamiliar user can attach and activate the device within 2 minutes using brief instructions	Usability testing has not yet been completed.	TBD
Fault Response	Detects a temperature-sensing fault, warns the user, and turns off the heater	Disconnected temperature-sensing connections during operation and observed the alarms and heater response.	PASS
Warm-Up Time	Outlet gas reaches 34°C within 10 minutes	Measured the time from heater activation until the gas first reached 34°C.	PASS - 8 min

Conclusion and Future Plans

Neo-Thermo is a low-cost retrofit device that adds oxygen warming and temperature monitoring to existing NICU infrastructure in resource-limited Ugandan hospitals, without requiring full system replacement

In the future, we aim to:

- Incorporate a battery** to ensure the device is not AC dependent
- Transition to a custom PCB** for greater durability and system integration
- Expand user testing** to evaluate long-term performance and real-world adherence.

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Footnotes and Sources: <https://tinyurl.com/vt3p5axr>